

Electricity Price Prediction Using Machine Learning

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Executive Summary This study investigates the use of machine learning methods to predict electricity prices using electricity consumption, generation sources, and time-based variables. The analysis was carried out in Apache Spark using PySpark on the Databricks platform. Two predictive models were implemented: Linear Regression and Random Forest Regression. Their performance was evaluated using Root Mean Squared Error (RMSE) and Coefficient of Determination (R^2). The results showed that Random Forest substantially outperformed Linear Regression, indicating that electricity price behaviour is better explained by nonlinear relationships than by a simple linear structure. Feature importance analysis further revealed that imported coal generation, solar generation, and electricity consumption were among the most influential factors affecting electricity price prediction.

I. Introduction

Electricity markets are among the most complex systems in modern energy infrastructure. Electricity prices fluctuate continuously due to multiple interacting factors such as energy demand, generation capacity, fuel costs, weather conditions, and market regulations. Accurate forecasting of electricity prices is therefore essential for energy producers, policymakers, and market operators, as it supports better decision-making, risk management, and efficient resource allocation.

Traditional forecasting approaches often rely on statistical models and time-series techniques. However, electricity price dynamics are frequently nonlinear and influenced by many interacting variables. As a result, conventional statistical methods may struggle to capture these complex relationships.

In recent years, machine learning techniques have become increasingly popular for electricity price forecasting. These methods can learn complex patterns from large datasets and identify nonlinear relationships between variables. Machine learning models have demonstrated strong predictive capabilities in various energy market applications.

The aim of this study is to develop machine learning models capable of predicting electricity prices based on electricity consumption, generation sources, and temporal variables. Two models were implemented in this project: Linear Regression and Random Forest Regression. Both models were developed using PySpark in the Apache Spark environment. Model performance was evaluated using Root Mean Squared Error (RMSE) and Coefficient of Determination (R^2) metrics. The models were then compared to determine which approach provides better predictive performance.

II. Literature Review

Electricity price forecasting has been widely studied in the fields of energy economics, forecasting, and machine learning. Because electricity prices are highly volatile and affected by multiple drivers, a wide variety of approaches has been proposed in the literature.

Conejo et al. examined day-ahead electricity price forecasting and highlighted the importance of modelling market structure and demand patterns within deregulated power systems. Their work established an important foundation for subsequent electricity price forecasting studies.

Weron provided one of the most influential reviews in the field and emphasised that electricity prices differ significantly from conventional financial or commodity series due to mean reversion, seasonality, volatility spikes, and non-storability. This makes electricity price forecasting particularly challenging and motivates the use of specialised methods.

Lago et al. reviewed the state of the art in day-ahead electricity price forecasting and concluded that machine learning and ensemble approaches frequently outperform simpler benchmark models in practical forecasting tasks. Their review also stressed the importance of rigorous evaluation and reproducible modelling pipelines.

In the broader machine learning literature, Breiman introduced Random Forests as a robust ensemble learning framework that performs well in nonlinear problems and provides interpretable feature importance measures. This makes Random Forest especially suitable for forecasting tasks involving structured tabular data.

Overall, the literature suggests that electricity price forecasting benefits from methods capable of handling nonlinear behaviour, interactions among variables, and significant volatility in the target variable. This study therefore adopts both a baseline linear model and a more flexible ensemble learning model.

III. Dataset Description

The dataset used in this study contains information about electricity generation sources, electricity consumption, and electricity prices. These variables make it possible to analyse how different generation sources and demand levels influence electricity price fluctuations.

The target variable in this study is electricity price measured in TRY/MWh. The explanatory variables include electricity consumption, total generation, energy generation from different sources, and temporal variables derived from the timestamp.

Before model development, the dataset was examined to ensure data quality. The following checks were performed:

- Missing value analysis
- Duplicate record detection
- Verification of negative values
- Data type inspection

These steps ensured that the dataset was suitable for machine learning analysis. A sample of the dataset is shown below.

Time	consumption_MWh	total_generation_MWh	natural_gas	coal_imported	wind	TRY/MWh	USD/MWh	EUR/MWh
01:01:2018:00:00	27412.81	27412.81	6926.78	5314.05	1179.51	207.6	55.04	45.97
01:01:2018:01:00	26324.39	26324.39	6603.44	5715.61	1148.46	205.34	54.44	45.47
01:01:2018:02:00	24635.32	24635.32	6040.77	5701.37	1216.38	164.94	43.73	36.53
01:01:2018:03:00	23872.12	23872.12	5938.46	5205.43	1298.11	154.52	40.97	34.22
01:01:2018:04:00	23194.89	23194.89	5665.41	4804.19	1220.91	112.64	29.86	24.95

Figure 1: Dataset sample preview.

IV. Data Preprocessing and Feature Engineering

Data preprocessing is an essential step in machine learning workflows. Properly prepared data can significantly improve model performance and ensure that the input variables are suitable for modelling.

In this project, several preprocessing and feature engineering steps were applied. First, the time variable was converted into a proper timestamp format so that temporal features could be extracted. Then, the following time-based variables were created:

- Hour of the day
- Month of the year
- Day of the week
- Weekend indicator

Electricity consumption and prices often vary significantly depending on time. For example, demand is typically higher during certain hours of the day and may differ between weekdays and weekends. Including temporal variables therefore allows the models to capture these systematic patterns.

In addition, PySpark's VectorAssembler was used to combine multiple feature columns into a single feature vector required by Spark machine learning models.

V. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed to better understand the relationships between variables and identify meaningful patterns in the dataset.

The first part of the analysis focused on electricity consumption behaviour. Summary statistics showed that the average hourly electricity consumption was approximately 35,166 MWh, with noticeable variation across observations. This confirmed that the dataset contained substantial fluctuations in demand over time.

The relationship between electricity consumption and electricity price was then examined. The visual evidence suggested that higher levels of electricity demand were generally associated with higher electricity prices. Although the relationship was not perfectly linear, the overall trend indicated that demand pressure plays a significant role in electricity price formation.

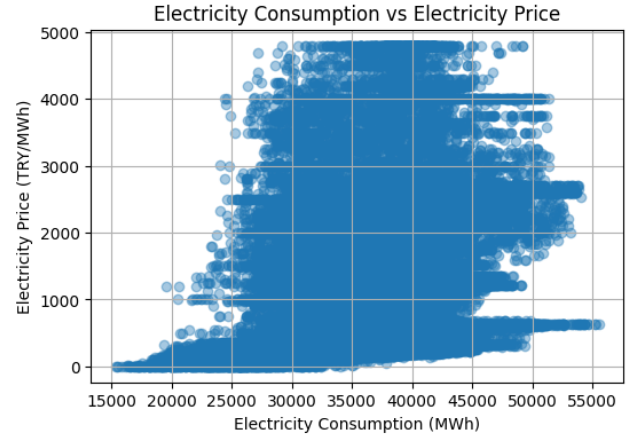


Figure 2: Relationship between electricity consumption and electricity price.

The daily consumption profile was also investigated. Average electricity demand varied systematically throughout the day, with lower demand during the early morning hours and higher demand during daytime and early evening periods. This pattern is consistent with expected human and industrial activity cycles.

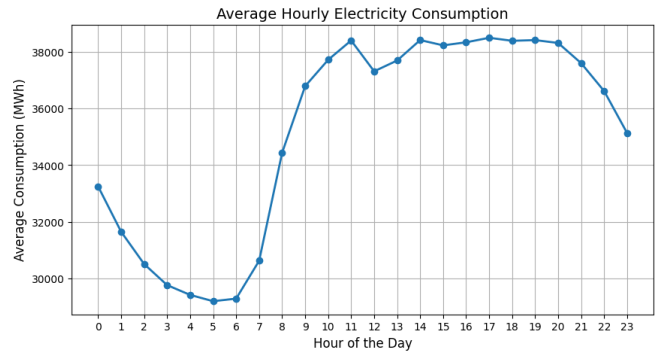


Figure 3: Average hourly electricity consumption pattern.

Monthly demand patterns revealed clear seasonality in electricity consumption. Demand levels were higher

during summer months and lower during spring months, indicating that electricity usage changes over the course of the year in a structured manner.

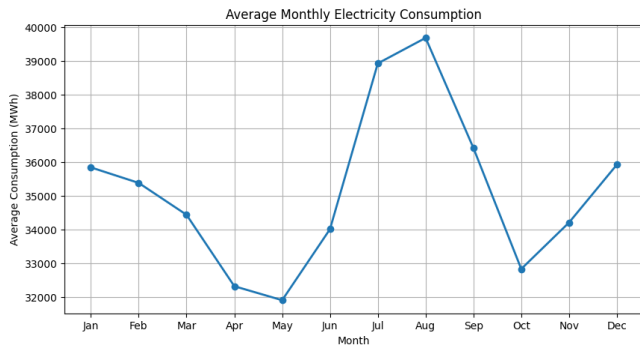


Figure 4: Average monthly electricity consumption pattern.

Electricity price distribution was also examined. The histogram showed a strongly right-skewed distribution, meaning that most prices were concentrated at lower levels while a smaller number of observations represented extreme price spikes. This is typical in electricity markets where most hours operate under normal conditions but rare stress periods can lead to sharp price increases.

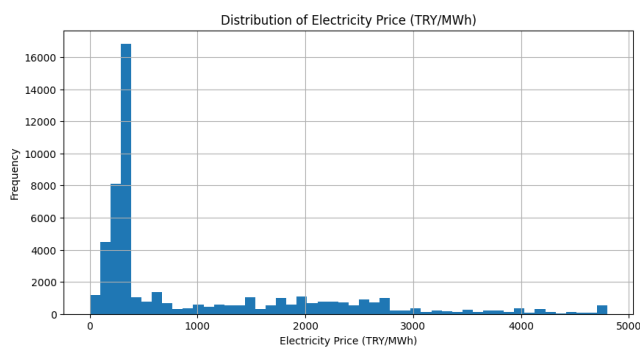


Figure 5: Distribution of electricity prices.

A boxplot of electricity prices further confirmed the presence of extreme outliers. These price spikes are consistent with the volatile nature of electricity markets and may be explained by temporary imbalances between demand and supply or changes in the generation mix.

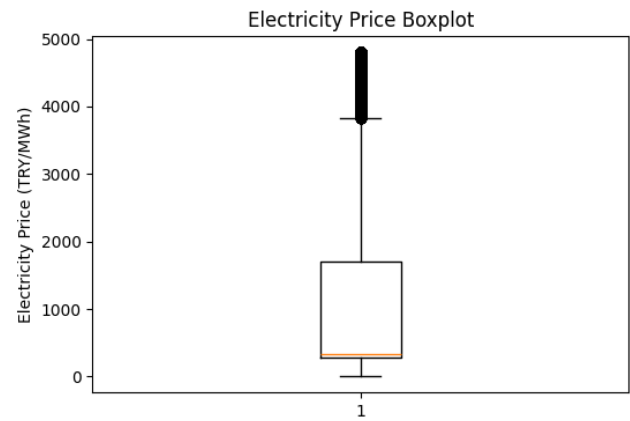


Figure 6: Boxplot of electricity prices showing extreme price spikes.

To better understand relationships between key explanatory variables and electricity prices, a demand-versus-price trend and a correlation matrix were produced. The results indicated a strong positive relationship between electricity demand and total generation, showing that production closely follows demand. A moderate positive relationship was also observed between electricity demand and electricity price, suggesting that higher consumption tends to push market prices upward.

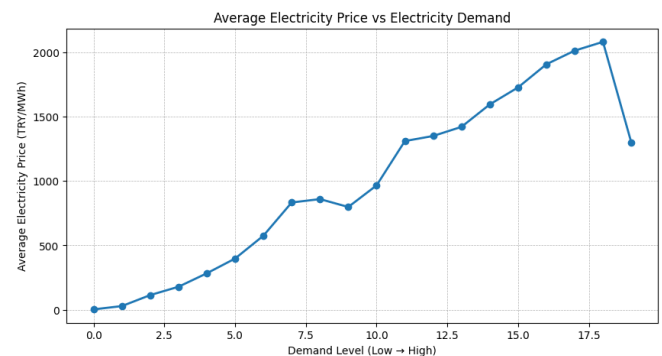


Figure 7: Relationship between electricity demand level and electricity price.

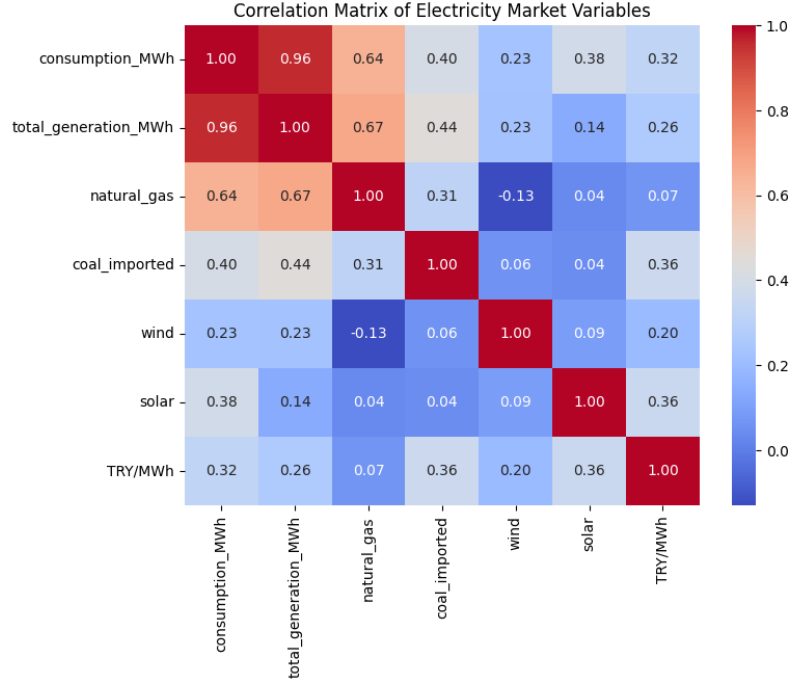


Figure 8: Correlation matrix of key electricity market variables.

VI. Machine Learning Models

Two machine learning models were implemented in this project to predict electricity prices.

A. Linear Regression

Linear Regression is a standard baseline model used to estimate linear relationships between a target variable and a set of explanatory variables. In this study, it was used as a benchmark model to assess how well a simple linear structure could explain electricity prices.

The model used the following inputs: electricity consumption, total electricity generation, natural gas generation, imported coal generation, wind generation, solar generation, hour, month, and day of week.

Although Linear Regression is easy to interpret, it is limited in its ability to capture nonlinear behaviour. Given the complexity of electricity markets, it was expected that this model might provide only moderate predictive performance.

B. Random Forest Regression

Random Forest is an ensemble learning method that combines multiple decision trees. It is particularly useful when the relationship between variables is nonlinear or when complex interactions exist between predictors.

Random Forest was selected because electricity prices are influenced by multiple interacting factors and these relationships are unlikely to be strictly linear. In addition, Random Forest provides feature importance scores, allowing interpretation of which variables contribute most strongly to prediction.

VII. Model Evaluation

The models were evaluated using two standard regression metrics:

- Root Mean Squared Error (RMSE)
- Coefficient of Determination (R^2)

RMSE measures the typical size of prediction errors, with lower values indicating better model performance. R^2 indicates how much of the variance in the target variable is explained by the model, with higher values representing better explanatory power.

The Linear Regression model produced relatively weak results. Its RMSE was high and its R^2 value indicated that only a limited proportion of the variation in electricity prices could be explained by a linear relationship. In addition, some of the generated predictions were unrealistic, including negative predicted prices, which further highlighted the limitations of the linear approach in this context.

By contrast, the Random Forest model performed considerably better. It achieved a lower RMSE and a substantially higher R^2 , indicating that it was better able to capture the nonlinear structure of the problem.

Model	RMSE	R2
Linear Regression	948.7316845	0.289269691
Random Forest	651.3138104	0.66503564

Figure 9: Performance comparison table for Linear Regression and Random Forest.

The model comparison clearly showed that Random Forest outperformed Linear Regression. This supports

the expectation that electricity prices are driven by non-linear relationships and interactions between demand, generation sources, and time-based factors.

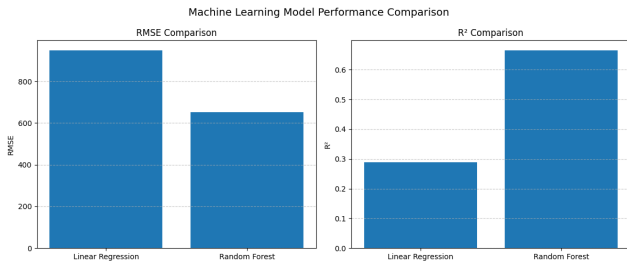


Figure 10: Graphical comparison of Linear Regression and Random Forest models.

VIII. Actual vs Predicted Analysis

To gain a deeper understanding of model behaviour, actual and predicted electricity prices were plotted for both models.

For Linear Regression, the actual-versus-predicted scatter plot showed a wide dispersion around the ideal prediction line. This indicated that the model struggled to track actual prices accurately, especially in more volatile price ranges.

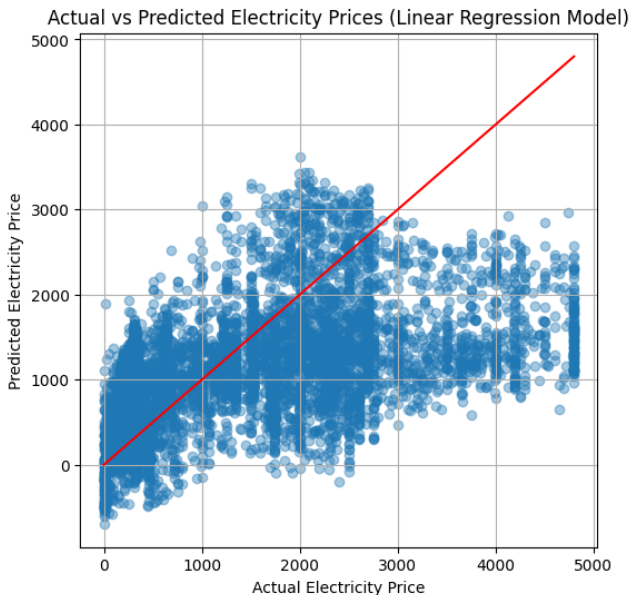


Figure 11: Actual versus predicted electricity prices using Linear Regression.

For Random Forest, the actual-versus-predicted scatter plot showed a noticeably better alignment with the ideal prediction line. While the model still exhibited some dispersion, particularly at higher price levels, the overall fit was substantially improved compared to Linear Regression.

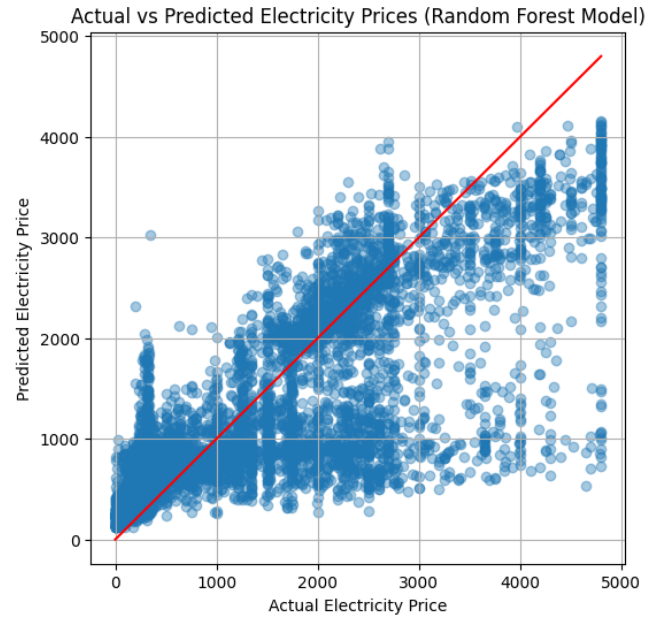


Figure 12: Actual versus predicted electricity prices using Random Forest.

These findings confirm that Random Forest provides a more realistic and reliable prediction structure for electricity price forecasting in this dataset.

IX. Feature Importance Analysis

One of the key advantages of the Random Forest model is its ability to estimate the relative importance of each input feature.

Feature	Importance
coal_imported	0.518891419
solar	0.195242759
consumption_MWh	0.06346023
natural_gas	0.049841495
month	0.045700844
wind	0.041894855
total_generation_MWh	0.041033777
hour	0.039240862
day_of_week	0.004693758

Figure 13: Feature importance values obtained from the Random Forest model.

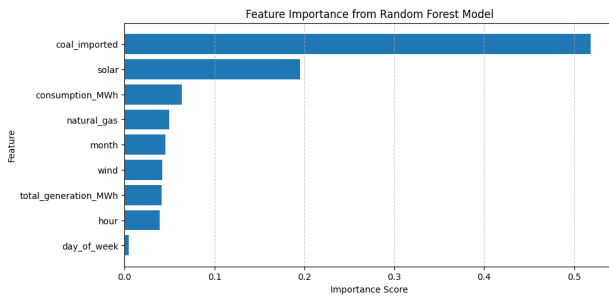


Figure 14: Feature importance chart from the Random Forest model.

The feature importance results indicated that imported coal generation was the most influential predictor, followed by solar generation and electricity consumption. This suggests that both the generation mix and market demand play major roles in determining electricity prices.

The prominence of imported coal generation may reflect the economic importance of fuel-based generation in market pricing. Similarly, the significance of solar generation suggests that changes in renewable energy contribution may also influence electricity market behaviour. Electricity consumption itself remained an important driver, which is consistent with the earlier exploratory findings.

X. Discussion

The results of this study demonstrate that machine learning models can effectively predict electricity prices using demand, generation sources, and temporal variables.

The comparison between Linear Regression and Random Forest is particularly informative. Linear Regression served as a useful baseline model, but its performance was limited by the assumption of linearity. Electricity prices are influenced by multiple nonlinear interactions, which the linear model could not fully capture.

Random Forest, on the other hand, achieved substantially better predictive performance. It was able to model complex relationships between electricity demand, generation sources, and price behaviour more effectively. The improvement in RMSE and R^2 indicates that Random Forest is considerably more suitable for this type of forecasting problem.

The feature importance analysis further strengthened the interpretation of the results by identifying the variables that contributed most strongly to prediction. This adds analytical value beyond pure forecasting accuracy, as it helps explain the underlying drivers of electricity price movements.

XI. Conclusion

This study developed and evaluated machine learning models for electricity price prediction using electricity consumption, generation sources, and time-based variables.

Two models were implemented: Linear Regression and Random Forest Regression. The results showed that Ran-

dom Forest significantly outperformed Linear Regression, achieving lower prediction error and higher explanatory power.

The exploratory analysis revealed meaningful relationships between demand, generation, and electricity price. Feature importance analysis further showed that imported coal generation, solar generation, and electricity consumption were among the most influential variables in predicting electricity prices.

Overall, the findings suggest that machine learning provides an effective framework for analysing and predicting electricity prices. The results also demonstrate that nonlinear methods such as Random Forest are better suited to electricity market forecasting than simple linear models.

Future work could extend this study by incorporating additional explanatory variables, such as weather conditions, fuel market data, or lagged time-series features. More advanced models such as Gradient Boosting or XGBoost could also be explored to further improve predictive performance.

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